



# LUNAR FARSIDE TECHNOSIGNATURE AND TRANSIENTS TELESCOPE (LFT3)

A Legacy Survey Mission



BREAKTHROUGH  
LISTEN

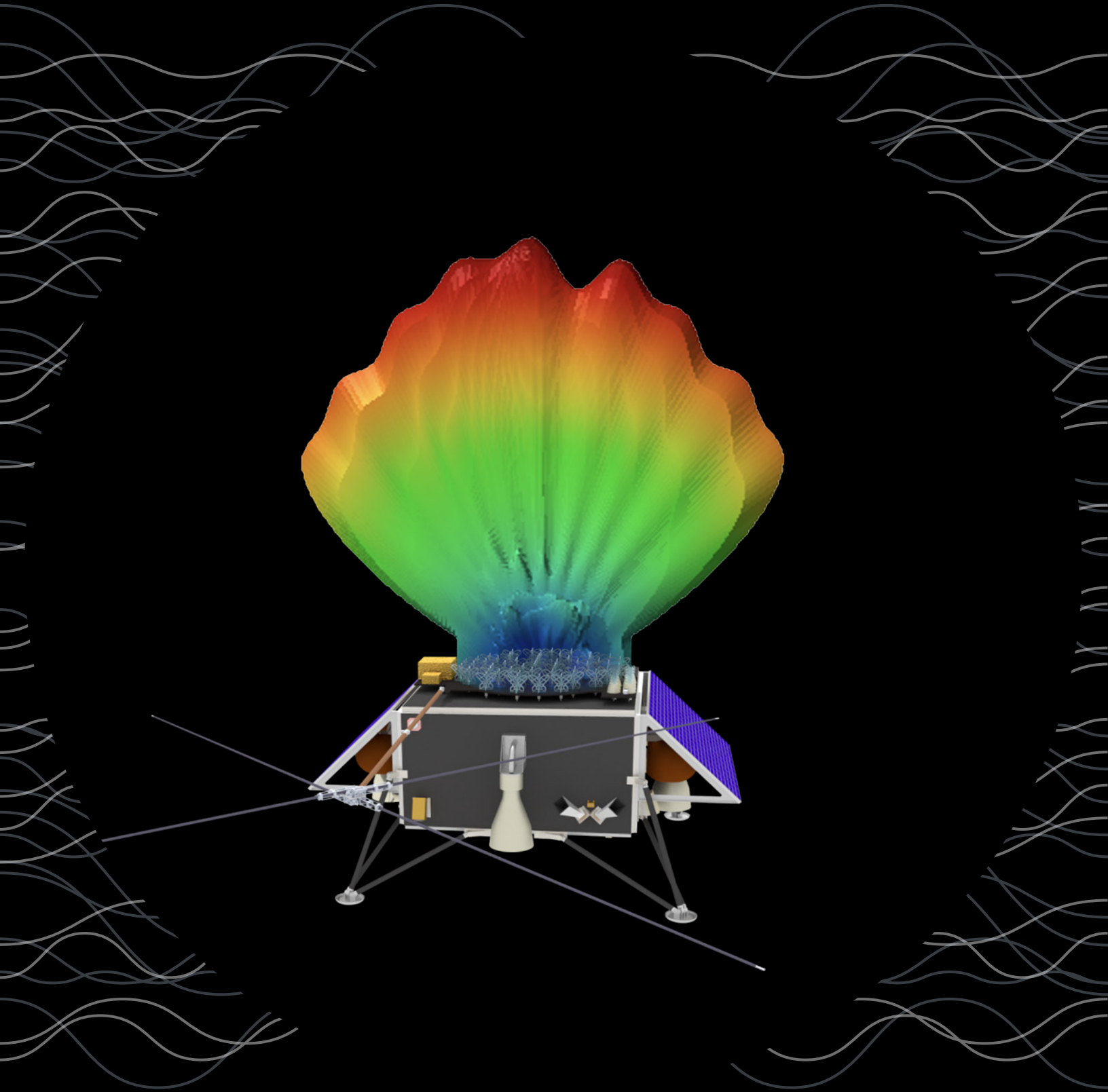
BREAKTHROUGH  
INITIATIVES

UNIVERSITY OF  
OXFORD

RADIO ASTRONOMY  
SYSTEMS | BYU

SSL

HELIOSPACE



UHF-Lo Multibeam Arrangement

## Radio Science Impact

Radio wavelengths provide the ability to observe and understand the roughly 90% of ordinary matter that comprises our Universe and is inaccessible to other wavelength bands. Radio wavelengths also uniquely allow us to observe and survey the entirety of the very early times of the Universe when the first structures began to coalesce.

**Radio wavelengths potentially provide our best chance to glimpse technological signals from other civilizations in our Universe.**

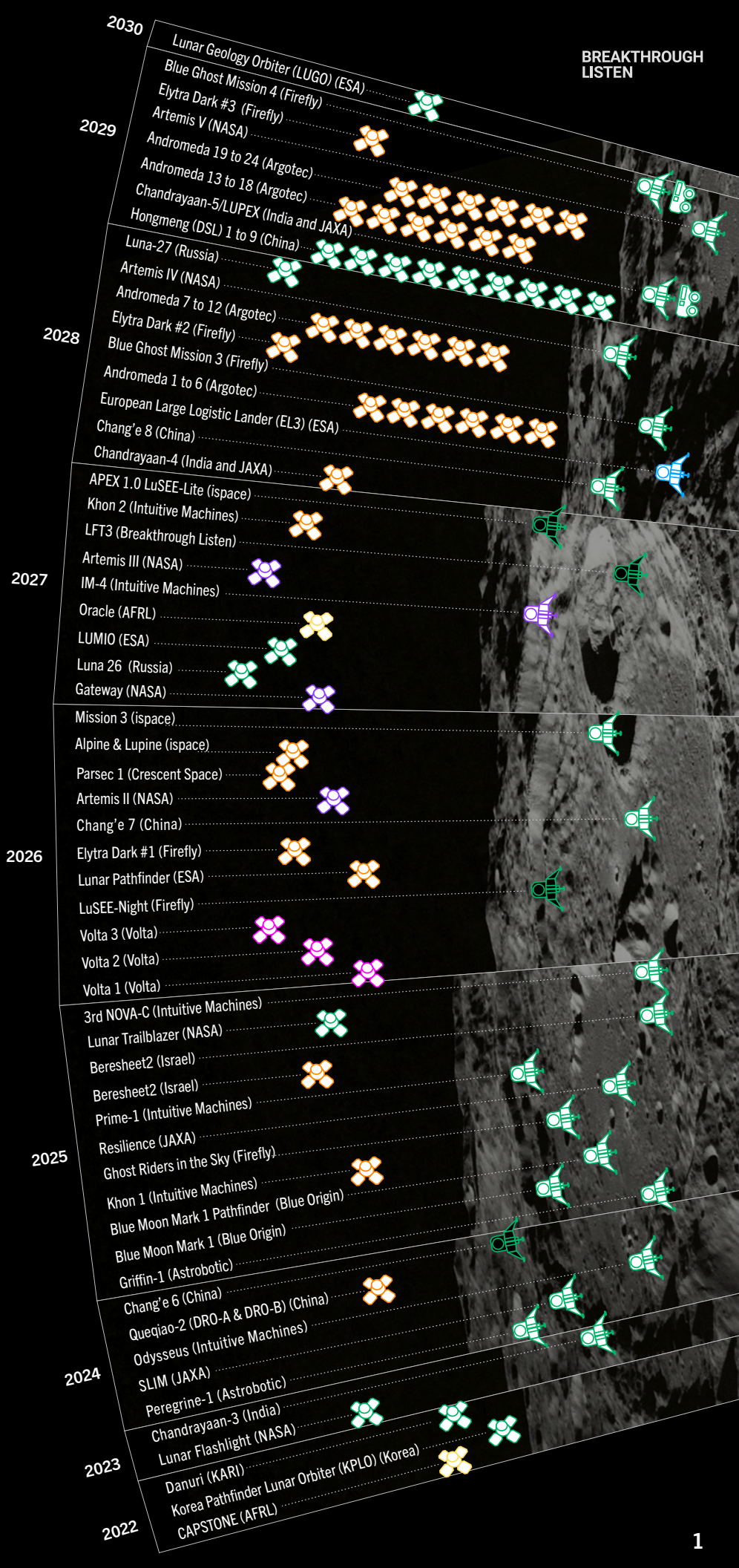
The lunar farside is a uniquely radio-quiet location in the inner solar system since it is always blocked from Earth's line-of-sight.

Humanity is poised to explore the cislunar environment, but has not yet arrived in numbers.

Now is the time to conduct this once-in-human-history survey of the quiet cosmic radio frequency environment.

By the end of this decade, many orbiters and landers will be present, impacting the pristine environment and humanity will lose this unique opportunity.

-  NAVIGATION
-  SCIENCE
-  COMMUNICATION
-  EXPLORATION
-  LOGISTICS
-  POWER BEAMING
-  FAR SIDE





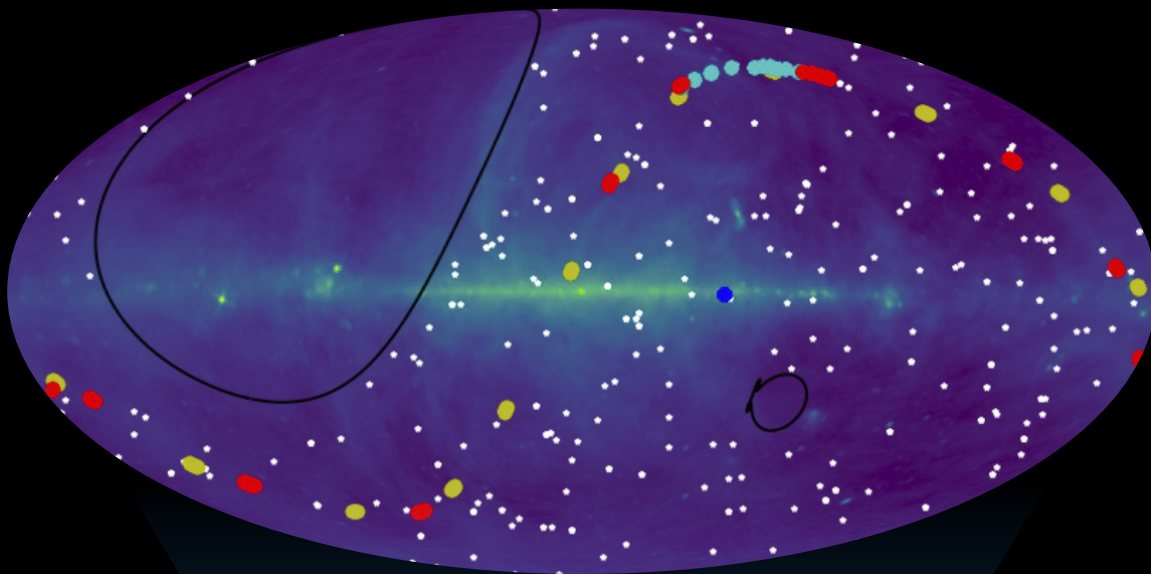
The 1973 measurements taken by NASA's RAE-2 mission highlighting the extreme radio-quiet lunar farside environment remain the exemplar, however recent missions (like China's 2018 Longjiang orbiters) are beginning to show this same promise.

China is the only country to land and operate from the lunar farside with their Chang'E 4 mission, which was corrupted by self-interference.

The full system provides sensitive frequency coverage from 1-2700 MHz.

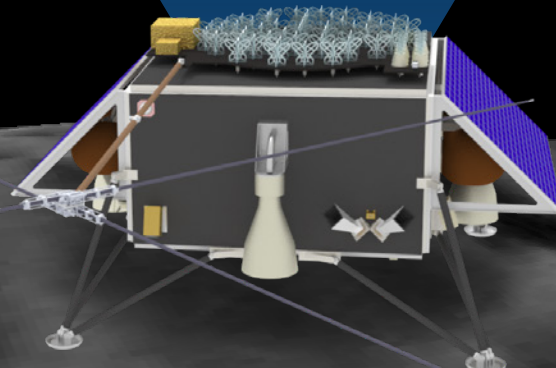
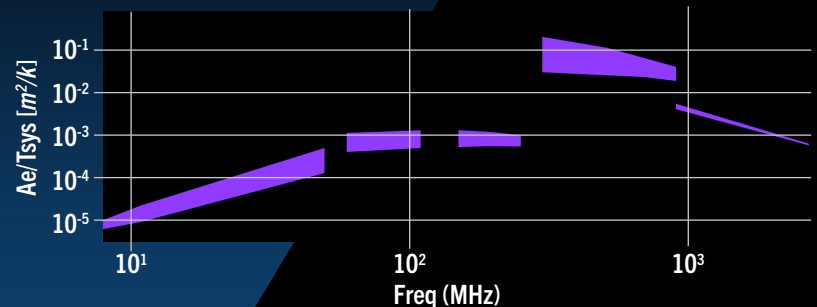
LFT3 builds on the NASA/DOE LuSEE-Night stand alone pathfinder in 2026 to land on the farside and use lander resources to provide continuous far side observations for 6 Earth years.

The LFT3 midband compact dual polarization multi-beam phased array provides wide sky coverage with full aperture sensitivity.



- MARS
- JUPITER
- SUN
- ALPHA CENTAURI
- NEAREST STARS

EFFECTIVE AREA / SYSTEM TEMPERATURE

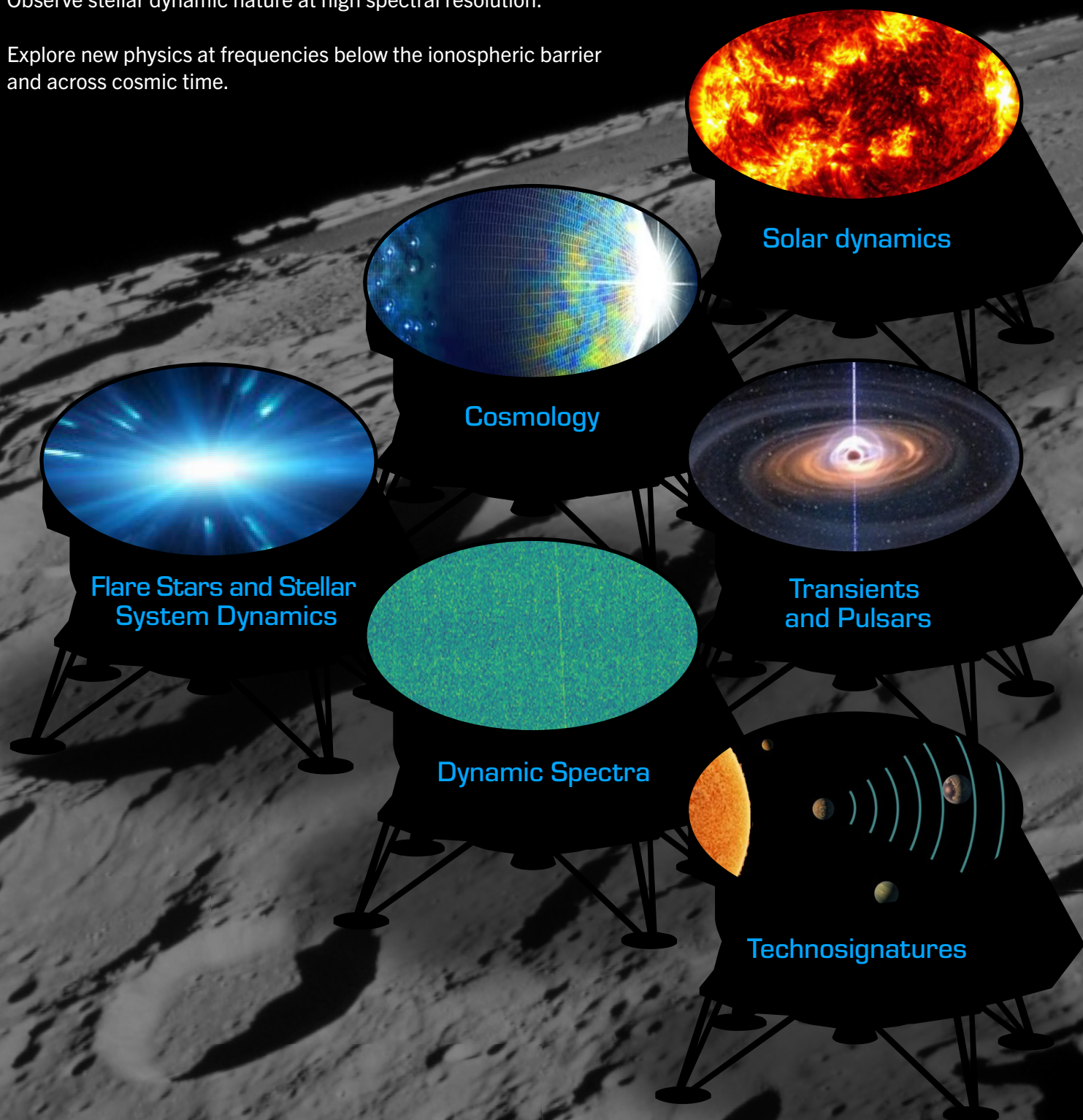


Search for extraterrestrial life beyond the sphere of human influence and across the radio band.

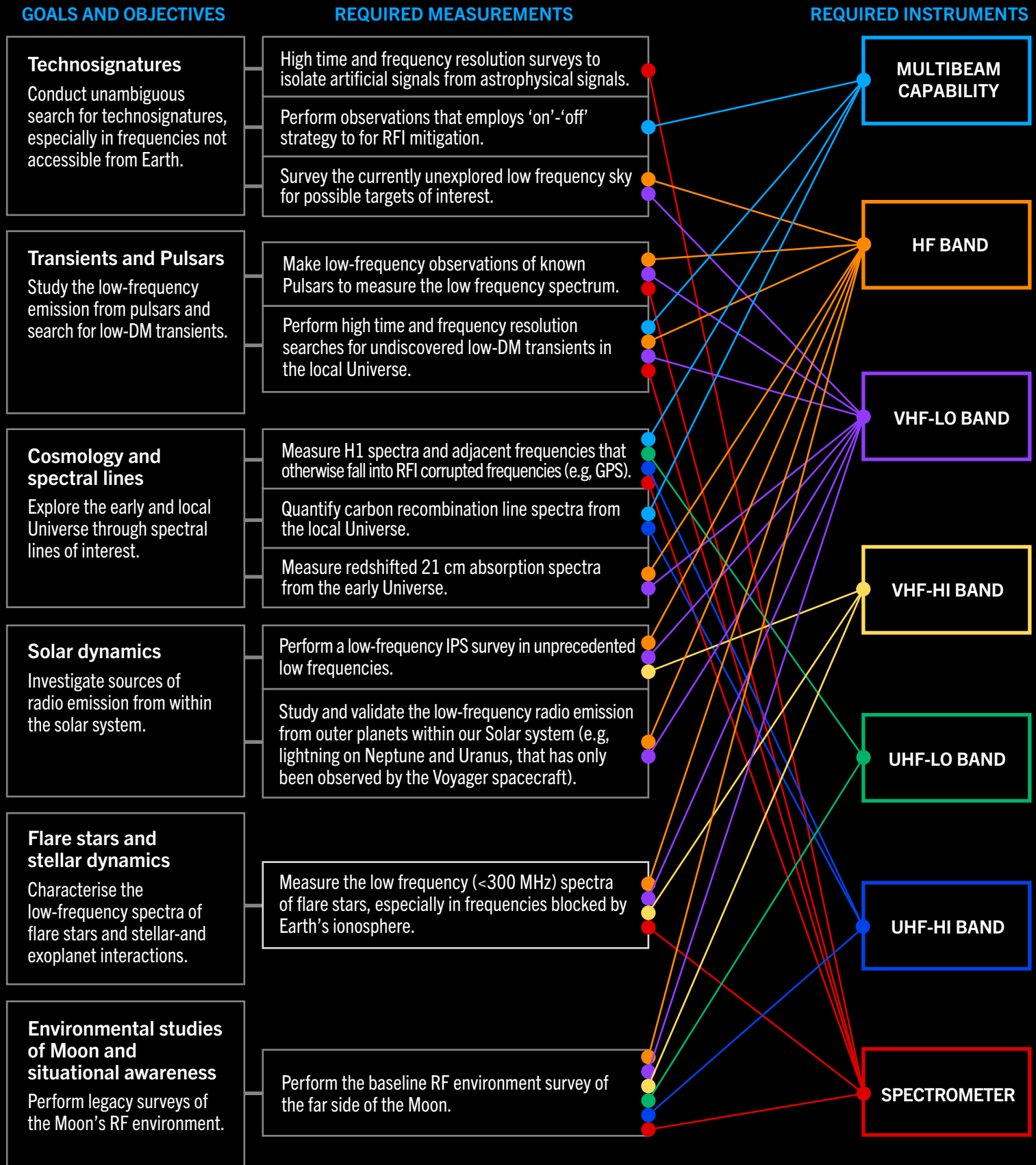
Study unique periodic or transient events across a wide range of wavelengths and timescales.

Observe stellar dynamic nature at high spectral resolution.

Explore new physics at frequencies below the ionospheric barrier and across cosmic time.



## Science Traceability Matrix

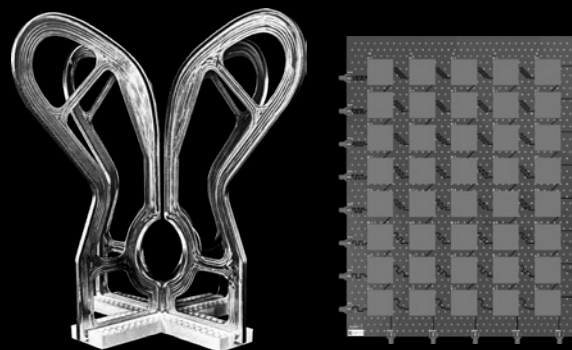




## Primary Instrumentation

### UHF-LO PHASED ARRAY AND BEAMFORMER

The LFT3 UHF-Lo compact dual polarization multibeam Vivaldi phased array provides wide sky coverage with full aperture sensitivity. Operating bandwidth is 300-900 MHz. The system incorporates a Blass matrix analog beamformer to produce 10 beams.

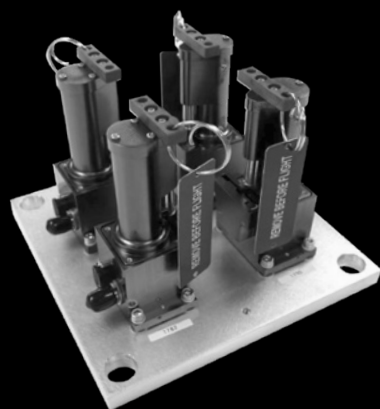
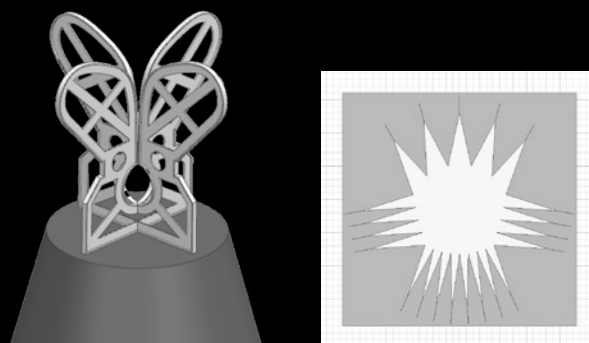


### SPECTROMETER

The flexible spectrometer allows for on-board dynamic spectra to be computed and stored along with triggered baseband data. It interfaces with the rest of the system to provide data to be transmitted back to Earth.

### UHF-HI PHASED ARRAY AND BEAMFORMER

The smaller UHF-Hi array operating in the 900-2700 MHz range uses elements 1/3 the size of the UHF-Lo array. The system incorporates a Rottman lens analog beamformer to produce 3 beams.

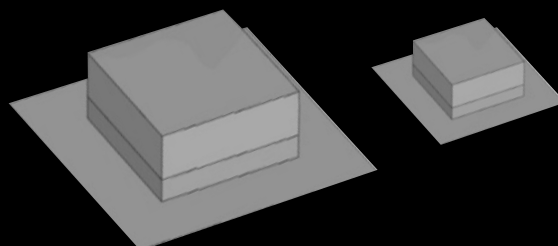


### HF ANTENNAS

The LFT3 HF Antennas operating in the 1-50 MHz range and are the same as those flown on LuSEE-Night and have heritage to the antennas for ROLSES.

### VHF-LO AND VHF-HI ANTENNA

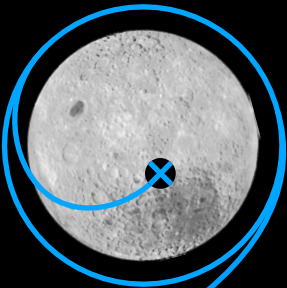
The VHF Antennas operating in the 60-110 MHz and 150-250 MHz range is a scaled concept based on the the Dark Ages Polarimeter Pathfinder (DAPPER) instrument.



## Getting to the Moon

Commercial Launch Vehicle, Lunar Lander, and Communication Relay vendors exist that have the capability to execute the LFT3 mission.

LANDING SITE



TERRAN R

FALCON 9

ECLIPSE



DEORBIT MANEUVER

BRAKING MANEUVER

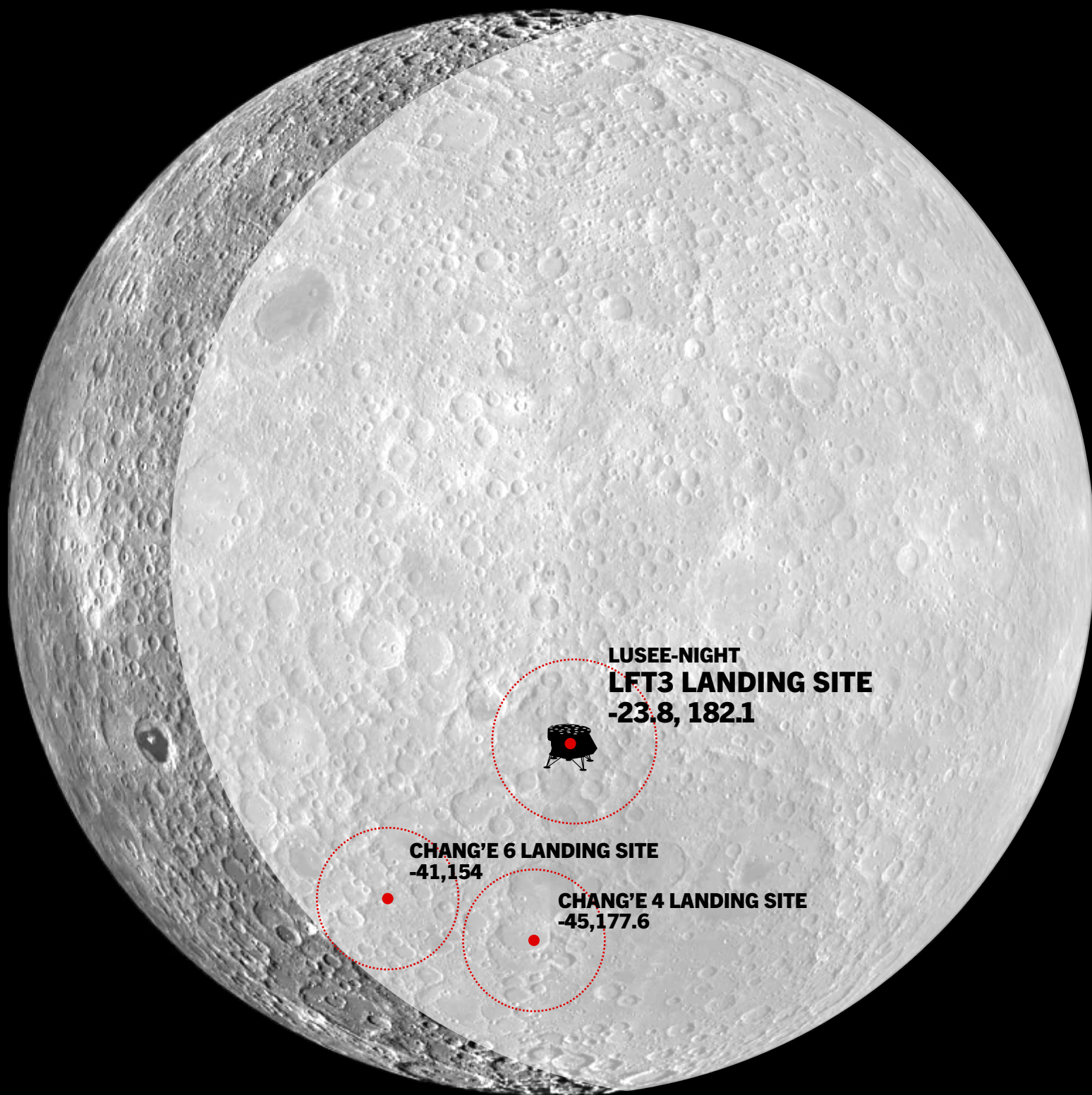
COMMUNICATION RELAY

FINAL DESCENT

| INITIAL DESIGN | DETAIL DESIGN | BUILD AND INTEGRATION | LAUNCH    |           |
|----------------|---------------|-----------------------|-----------|-----------|
| Start          | 6 Months      | 18 Months             | 30 Months | 36 Months |



For more information contact **David DeBoer**  
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**LUSEE-NIGHT  
LFT3 LANDING SITE  
-23.8, 182.1**



**CHANG'E 6 LANDING SITE  
-41,154**

**CHANG'E 4 LANDING SITE  
-45,177.6**